Ghofrane Mahfoudhi

ICT Engineer | Specialized in AI • Generative Models • Agentic Systems

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Summary

Information and Communication Technologies (ICT) Engineer, recently graduated from **SUP'COM**, passionate about **Artificial Intelligence**, from **Deep Learning** to **Generative AI** and **Agentic AI**. Interested in designing and developing AI-powered systems across different domains, with a focus on applying AI to solve real-world problems. With international experience in **Tunisia and Canada**, I enjoy exploring new technologies, working in diverse environments, and taking on new challenges

Education

- **Higher School of Communications of Tunis, SUP'COM** — *2023 - 2026*
  ICT Engineering — Ariana, Tunisia
- **Preparatory Institute for Engineering Studies of Sfax** — *2021 - 2023*
  Preparatory Classes in Physics and Chemistry - Ranked 24th/1500 in the National Engineering Entrance Exam — Sfax, Tunisia
- **Hedi Soussi High School** — *2020 - 2021*
  Scientific Baccalaureate (Highest Honors) — Sfax, Tunisia

Research & Professional Experience

- **Polytechnique Montréal – SEMLA Lab** — *March 2026 – Present*
  AI Research Intern — Montreal, Canada
  - **Policy-Faithful Framework for Evolving Documents:** Designed and developed a framework to improve the reliability of **Retrieval-Augmented Generation (RAG)** systems for evolving policy documents. The framework identifies meaningful changes across document versions, organizes information into precise evidence units, and enables version-aware and traceable evidence retrieval to support reliable AI-generated answers. The project aimed to reduce outdated or incomplete information and improve the **reliability, faithfulness, and traceability** of generated answers.
  - **Technologies:** Python, LLMs, RAG, Hugging Face, Sentence Transformers, PyTorch, Weaviate, DeepEval, GEval, FastAPI, HTML, CSS, JavaScript, Docker, Git
- **Converty (Fintech Startup)** — *September 2025 – January 2026*
  AI / Data Intern — Tunis, Tunisia
  - **Automated Advertising Campaign Monitoring System:** Designed and developed a platform to automate the monitoring and analysis of **Converty clients' Facebook advertising activity**. The system identifies Facebook pages associated with e-commerce domains, collects active advertisements, and determines whether they redirect to **Converty or competing platforms**, providing analytical reports to support client monitoring and strategic decision-making.
  - **Technologies:** Python, Facebook Ads Library, Apify, MongoDB, Streamlit, Docker
- **KPIT Technologies** — *July 2025 – August 2025*
  AI / Data Intern — Sfax, Tunisia
  - **Automated Data Engineering Pipeline with LLMs AI Agents:** Designed and developed a system to automate key **data engineering tasks**, including the generation of **SQL queries** and **Grafana dashboards** from natural language descriptions. The solution reduces manual effort in data extraction and visualization, helping data engineers streamline repetitive tasks and focus on more complex analytical challenges.
  - **Technologies:** Python, LangChain, LLMs, SQL, SQL Server, Grafana API, Prompt Engineering, AI Agents
- **Numeryx MEA** — *July 2024 – August 2024*
  Software Development Intern — Sfax, Tunisia
  - **Car Park Management Web Application:** Designed and developed a web application to facilitate the management of **customers and vehicles** in a car park. The system enables employees to securely register, update, and manage customer and vehicle information through a centralized platform, improving data organization and access control.
  - **Technologies:** HTML, CSS, Bootstrap, PHP, MySQL, XAMPP

Key Projects

- **AI-Based Human Life Detection System for Rescue Operations** — *January 2025 - June 2025*
  - Designed and developed an **AI-based system** to detect signs of life in people trapped under rubble by combining **radar and thermal data**. The system analyzes subtle human movements and heat signatures to distinguish between living and non-living individuals, supporting more reliable victim detection in challenging disaster environments.
  - **Technologies:** Python, MATLAB, TensorFlow, Deep Learning, CNN, 3D-CNN, FMCW Radar, Thermal Imaging
- **EduBot – Intelligent Learning Assistant** — *November 2024 - December 2024*
  - Developed a **web platform** with an intuitive interface for educational content, integrated with an **AI chatbot** for course-related questions.
  - **Technologies:** Java Spring Boot, Angular, PostgreSQL, AI API

Skills

Artificial Intelligence & Machine Learning: Deep Learning, Generative AI, Agentic AI, LLMs, RAG, AI Agents, Sentence Transformers, PyTorch, TensorFlow, Prompt Engineering, OCR
Programming Languages: Python, C++, MATLAB, Java, PHP, SQL
Web Development: HTML, CSS, JavaScript
Backend & APIs: FastAPI
Databases & Vector Databases: MySQL, PostgreSQL, SQL Server, MongoDB, Weaviate
Cloud: AWS, Microsoft Azure
Data & Visualization: Excel, Power BI, Grafana
Tools: Docker, Git

Certifications

- **AWS Academy Graduate - Machine Learning Foundations** *2025*
- **CCNA: Introduction to Networks (Cisco Networking Academy)** *2024*

Volunteering

- **SEMLA 2026 - Engineering Trustworthy and Cost-Effective AI Multi-Agents Conference** *2026*
- **SUP'COM Entrepreneurial Challenge - Project management including market research and financial analysis** *2025*
- **PACTE PROJECT - Community project management for improving high school infrastructure** *2024*
- **SUP'COM Forum 2023 - Co-Head of Design** *2023*

Additional Information

Languages: Arabic (Native), English (B2), French (B2)